

3. Git- Branching and Merging

Experiment Name: Git Branching and Merging

GitLab Link:

<https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project>

Implementation

Step 1: Create a New Branch

Command

git branch GitNewBranch **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git branch GitNewBranch
```

Step 2: List All Available Branches

Command

git branch **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git branch
  GitNewBranch
* main
```

Step 3: Switch to the New Branch

Command

git checkout GitNewBranch **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
```

Step 4: Create a File in the Branch

Command

echo "Branch Content" > branch.txt **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitNewBranch)
$ echo "Branch Content" > branch.txt
```

```
deshm@Deshma MINGW64 ~/GitDemo (GitNewBranch)
$ ls
README.md  branch.txt  error.log  log/  welcome.txt
```

Step 5: Add File to Staging Area

Command

git add branch.txt **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the ne
xt time Git touches it
```

Step 6: Commit Changes in Branch

Command

git commit -m "Added branch file" **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch file"
On branch GitNewBranch
nothing to commit, working tree clean
```

Step 7: Check Git Status

Command

git status

Output

```
deshm@Deshma MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean
```

Merging

Step 8: Switch to Master Branch

Command

git checkout master **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

Step 9: Compare Master and Branch

Command

git diff master GitNewBranch **Output**

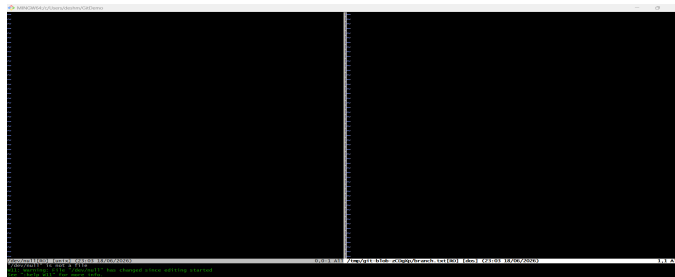
```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git diff main GitNewBranch
```

Step 10: View Visual Differences Using P4Merge

Command

git difftool master GitNewBranch

Output



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Step 11: Merge Branch with Master

Command

git merge GitNewBranch **Output**

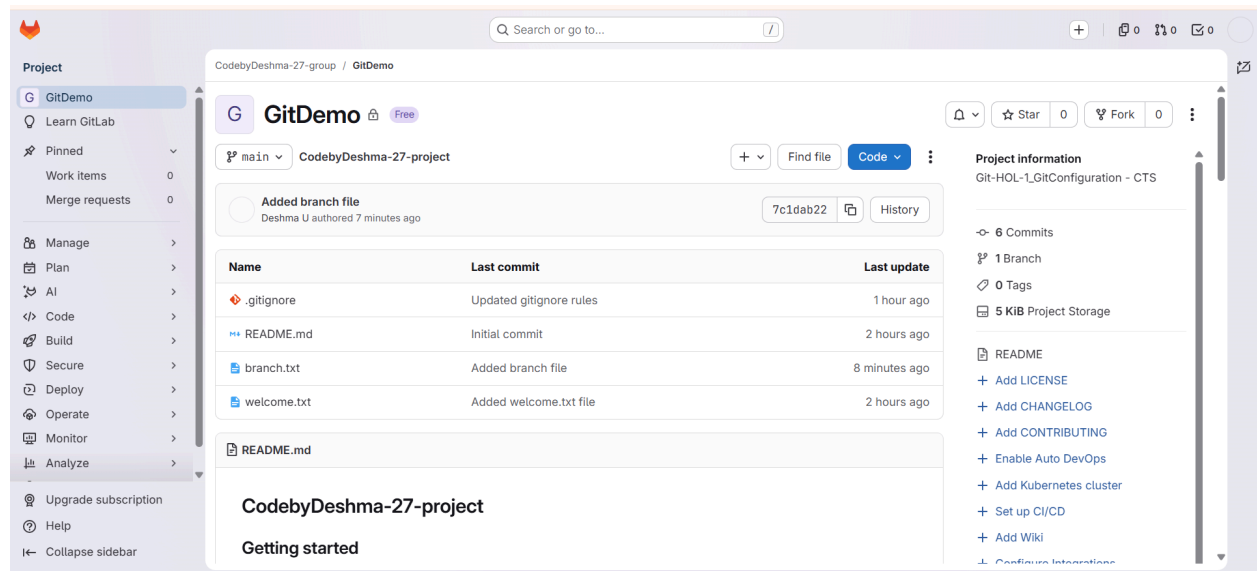
```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git merge GitNewBranch
Already up to date.
```

Step 12: View Commit Logs

Command

git log --oneline --graph --decorate **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
* 7c1dab2 (HEAD -> main, origin/main, GitNewBranch) Added branch file
* daa7ce7 Updated gitignore rules
* f151e69 Added gitignore file
* ccfbf47 Merge branch 'main' of https://gitlab.com/codebydeshma-27-group/Code
byDeshma-27-project
| \
| * 474fda1 Initial commit
| * 69c13cc Added welcome.txt file
```



Step 13: Delete Branch After Merging

Command

git branch -d GitNewBranch

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Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 7c1dab2).
```

Step 14: Final Git Status

Command

git status

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
~
```

Result

Thus, the Git branching and merging operations were performed successfully.